Strengths and Novelty of the Project

Investigation of Genes Causing Male Infertility Using Bioinformatics Techniques

A verification study of the roles of selected genes in male infertility using bioinformatics software, with re-execution by several artificial intelligence tools for classification, verification and comparison, and the design of a prototype website hosting an evidence database of genes mediating male infertility events.

Ali Saleh, B.Sc. Biomedical Sciences, Damascus University

1. Objective and findings

Objective. Non-obstructive azoospermia is genetically unexplained in most cases, and the bottleneck is interpreting variants rather than finding them. This computational study asks whether variants labelled “pathogenic” in 16 male-infertility genes are registered for male infertility or for other diseases, and which carry the strongest integrated evidence. Companion aims were to verify published information, to measure AI-tool performance quantitatively, and to build a prototype evidence-database website.

Findings. Of 712 pathogenic ClinVar variants, only 177 (24.9%) were registered for male infertility. All 17 XRCC2 variants were registered for other conditions, whereas TEX11, TEX14, SYCP2, M1AP and TERB1 variants were all registered for infertility. A “highest CADD” rule alone would have picked 6 of 15 representatives registered for other diseases. The final set comprises 15 representatives (8 stop-gain, 5 splice, 2 missense; CADD 27.2–46) and stayed stable across all nine threshold scenarios.

2. Design

A retrospective computational study on public data (GRCh38), with no patient samples. It began with 37 candidate genes from the Male Fertility Gene Atlas (MFGA), then required moderate-or-higher evidence in Houston 2022 and Zhao 2025 (HORMAD1 was excluded; MEI1, MEIOB and XRCC2 were added), leaving 16 genes. Gene–disease validity, variant pathogenicity and variant priority were kept as separate concepts. A gene enters only with moderate-or-higher ClinGen evidence, and a variant only if it is pathogenic or likely pathogenic with at least one submission linking it to the infertility phenotype. Groups and thresholds (CADD ≥ 20, gnomAD ≤ 0.001) were defined before results were examined, and sensitivity was tested across nine threshold scenarios.

3. Aim and importance

The project reverses the usual question: instead of asking “is there a new variant?”, it asks whether the variants we call pathogenic are really linked to male infertility. It re-examines variants already submitted to public databases and interprets them within the male-infertility context, as an interpretive layer that complements molecular diagnosis.

The need is documented. Houston et al. reported that only about 4% of infertile men receive a genetic diagnosis while 60–70% remain unexplained. Lillepea et al. (2024) found a molecular diagnosis in about 12% (64/521). Zhou et al. (2025) estimated the pooled exome yield in non-obstructive azoospermia at about 15% (95% CI 10–20%; nine studies; 1,728 individuals; low certainty). Riera-Escamilla et al. (2025) suggested that using two classification approaches shows earlier studies may have overestimated diagnostic yield, highlighting the need for a unified, infertility-specific classification. Oud et al. (2019) found only 78 genes confidently linked to male infertility out of 521 gene–disease relationships assessed. This is the space the project addresses.

4. Rarity

To our knowledge, no comparable study combines these elements. Claims are worded conservatively, because broad “first” claims would be incorrect: male-infertility databases exist, in silico missense studies exist, and AI benchmarking frameworks exist. The rarity lies in the combination of elements and in several of them individually.

- **Disease matching at the level of each ClinVar submission (SCV).** One variant can be registered for both spermatogenic failure and premature ovarian failure. This is the core novelty.
- **A reversed question.** Are “pathogenic” variants linked to male infertility, rather than ranked by predicted damage?

- **Gene strength does not transfer to every variant.** AR, a gene with top-level evidence for infertility, had only 7 of 385 variants registered for infertility in one project table, and about three quarters of “pathogenic” variants would be misattributed. XRCC2 had no disease-matched representative.
- **A striking quantitative result.** Only 24.9% of pathogenic variants in established male-infertility genes are registered for male infertility. This may prevent attributing a cancer-related or androgen-related variant to infertility.
- **A practical test of the “highest CADD” rule.** It would have selected 6 of 15 representatives registered for other diseases, because the score measures general damage, not relevance to a specific disease.
- **A paired comparison of six AI tools** on a complete candidate-filtering plan for a specific rare disease, against a reference verified from primary sources, with SHA-256 auditing showing that 21 of 22 outputs from two tools were identical. To our knowledge, we found no comparable design.
- **An interactive bilingual offline website** that recomputes the tests and lets readers re-apply the rule with their own thresholds.

5. Core novelty: disease matching at the submission level

The project does not rely on ClinVar’s aggregate classification. It descends to each submitted assessment (SCV) and asks for which disease the submitter registered the variant, male infertility or another disease. This is necessary because a single variant may be registered for spermatogenic failure and premature ovarian failure, as documented in ClinVar.

Numbers reported in the project.

- From 6,157 ClinVar variants in the candidate genes, 726 were pathogenic or likely pathogenic, and 712 fell in the 16 validated genes. Of the 712, only 177 (24.9%) were registered for male infertility; about three quarters were not.
- In another analysis with a denominator of 734 variants, 175 (23.8%) had any evidence of an infertility association and 477 (65.0%) were registered for other diseases only.
- In the first five-gene route, 38 variants matched infertility (6.6%), 11 were mixed, 466 were non-matched (80.5%) and 64 were unspecified.

Gene strength does not transfer to every variant.

- AR (the gene with the highest evidence): 7 of 385 variants (1.8%) were registered for infertility in one project table; in another table, 340 of 397 pathogenic variants (85.6%) were registered for other diseases only, chiefly androgen insensitivity syndrome and Kennedy disease.
- XRCC2: all 17 variants (17/17) were registered for other diseases, 15 of them linked to cancer predisposition, so it had no disease-matched representative.
- In contrast, all variants of TEX11, TEX14, SYCP2, M1AP and TERB1 were registered for infertility. The rule does not generalize one judgment to all genes; it shows which genes actually contribute to the infertility phenotype.

This is consistent with the literature on public databases. Shah et al. (2018) found apparent inflation in pathogenic variant sets relative to disease prevalence, reaching 11.5% of disorders in pathogenic sets and 92.3% in the conflicting-interpretation set. Other studies addressed the causes of conflicting interpretations in ClinVar (Lazareva et al. 2024) and the high proportion of downgrades when common variants are reinterpreted (Xiang et al. 2020).

“Highest CADD alone” is not enough. Choosing the highest CADD score alone would select 6 of the 15 representatives (40%) from variants registered for other diseases, while moving the CADD threshold from 15 to 25 changed none. In the missense category, 4 of 14 highest-CADD selections were not registered for male infertility; in NR5A1, a variant with CADD 35 was excluded because its registered disease was a mixed syndrome. The score measures general probability of harm, not relevance to a specific disease, and this appeared in the project’s own data.

6. How it differs from published work

To our knowledge, published studies in this field address related but different questions, and none combines the elements of this project.

- **Gene-validity reviews** (Oud 2019; Houston 2022; Zhao 2025) assess whether a gene causes male infertility. This project goes one level down and asks whether each variant is registered for male infertility, which these reviews do not examine.
- **Diagnostic-yield studies** (Lillepea 2024; Zhou 2025; Riera-Escamilla 2025) sequence patients and estimate how often a cause is found. This project uses no patient samples and claims no yield; it interprets variants already submitted to public databases. It is complementary, and it addresses the concern raised by Riera-Escamilla that inconsistent variant classification may inflate yield.
- **ClinVar reinterpretation studies** (Shah 2018; Xiang 2020; Lazareva 2024) examine misclassification and conflicting interpretations in general, not in male infertility, and not by checking which disease each submission was registered for.
- **Male-infertility databases** (Male Fertility Gene Atlas, Male Infertility Knowledgebase, IDDB, FertilityOnline, SpermatogenesisOnline) already integrate genes, expression and variants. Adding expression layers is not the novelty. This project adds disease matching at the level of each ClinVar submission, traces every value to its raw source through two independent routes, and lets readers re-apply the selection rule with their own thresholds.
- **AI benchmarks** (ClawBench) evaluate language models on ACMG/AMP classification in general. This project evaluates six AI tools on a complete candidate-filtering plan for one specific rare disease, against a reference verified from primary sources, with paired inferential tests (Cochran’s Q, exact McNemar with Holm correction) and SHA-256 auditing of output independence.

7. Methodological rigor

- Groups were defined before looking at results: H (highest CADD, no disease filter), C (highest CADD among disease-matched variants) and M (strongest missense variant per gene). This resembles pre-registration of a hypothesis and prevents post-hoc selection.
- Nine threshold scenarios (CADD 15, 20, 25; gnomAD 0.0001, 0.001, 0.01) all gave the same 15 representatives; thresholds change the size of the candidate list, not its identity.
- The rarity threshold does not create the list: 549 of 727 variants were absent from gnomAD, and the threshold excluded only one variant (MEI1, frequency 0.002171).
- Gene–disease validity, variant pathogenicity and variant priority were kept as three separate questions, so gene quality is not confused with variant quality and pathogenicity is not confused with disease association.
- Pre-definition and sensitivity testing are uncommon in graduation projects; they separate an analysis that looks good from one that is reproducible and testable.

8. Technical strength and traceability

- **Cell-to-source traceability:** a 48-sheet Excel evidence matrix whose values are pulled from source sheets with INDEX/MATCH formulas, so no value is copied by hand.
- **Expression recomputed from raw files** (GTEx v8 and HPA) exposed and corrected documented errors: MSH5 in testis from 3.10 to 34.60, and AR from 32.40 to 7.02.
- **Reproducibility infrastructure:** SHA-256 fingerprints, a rerunnable Colab notebook, read-only inputs, no overwriting of earlier outputs, every correction saved in a new folder, and no substitute value on failure.
- **Reference control:** MANE Select transcripts were used and the amino acid was checked in structural models. A numbering shift was detected in TEX11 and SHOC1, and SHOC1 was left uninterpreted rather than given a wrong interpretation.
- **Thirteen evidence layers in one matrix:** ClinVar (SCV), CADD v1.7, gnomAD v4.1.1, SpliceAI, REVEL, AlphaMissense, LOFTEE, NMD, 1000 Genomes, OMIM, GTEx v8, HPA (RNA and immunostaining), and AlphaFold/UniProt.

- **Missing values in three states** (not observed, technically unavailable, not applicable), with no imputation or inferred coordinates.
- **A declared bias handled:** because a “highest score” rule favours stop-gain variants, an expansion for the strongest missense variant per gene was added.

9. Data sources and tools

- **Genomic layer:** ClinVar (SCV level), CADD v1.7, gnomAD v4.1.1, 1000 Genomes.
- **Expression layer:** GTEx v8 and HPA, recalculated from raw files.
- **Functional layer:** SpliceAI, Pangolin, REVEL, AlphaMissense, LOFTEE, NMD prediction (290 of 342 variants, 84.8%), OMIM.
- **Structural layer:** AlphaFold models with pLDDT values, visualised in ChimeraX and PyMOL (four images per gene).
- **Analysis platforms:** Galaxy with VEP, and Python on Colab.
- **Statistics:** Cochran’s Q, exact McNemar with Holm correction, Wilson intervals, Jaccard, kappa, and a gene-level cluster bootstrap.

10. Missense variants, structure and NMD

- Because the selection rule favours high-scoring stop variants, an expansion selected the strongest missense variant per gene in two versions: highest CADD alone, and CADD with disease matching.
- About 13 missense variants were displayed in ChimeraX and PyMOL over AlphaFold models with pLDDT values, together with reproducible command files. This is a descriptive, hypothesis-generating analysis.
- NMD was predicted for all eight stop-gain representatives; this is a computational prediction, not functional proof.
- The final set contains 15 representatives in 16 genes (8 stop-gain, 5 splice, 2 missense), with CADD scores from 27.2 to 46; six are absent from gnomAD and the other nine have frequencies no higher than 0.00007.
- The most frequent genes in Zhou (2025), such as AR, TEX11, M1AP and DMRT1, overlap with the project set, and NR5A1 was the most frequent in Lillepea (2024). This is external consistency, not a product of the rule alone.

11. Reproducibility and statistical strength

- The plan was run through two independent routes, Galaxy with VEP and Python on Colab, and core values matched exactly (469/469 CADD scores, 131/131 allele frequencies). The unified analysis reproduced all 15 representatives.
- Tests were chosen to suit paired data: Cochran’s Q for overall comparison, exact McNemar for pairwise comparisons, Holm correction for multiple comparisons, Wilson intervals for proportions, the Jaccard index and kappa for agreement, and a gene-clustered bootstrap with 3,000 resamples (seed 20261012).
- Cochran’s Q for four tools: H, $Q = 9.790$, $p = 0.0204$; C, $Q = 7.828$, $p = 0.0497$; M, $Q = 14.087$, $p = 0.0028$. For three tools in M, $p = 0.0023$.
- Of 18 pairwise comparisons, four had a raw $p < 0.05$ and none survived Holm correction (smallest adjusted $p = 0.1143$). This was reported openly rather than hidden; the likely explanation is the small sample size (20–32 variants per group) combined with multiple comparisons.
- Tools were not ranked by “clinical accuracy” because no independent gold standard exists. They were ranked by one declared criterion: agreement with the documented internal reference.

12. Working with AI: a methodological contribution

- Six tools and models received the same plan: Claude (Opus 5.5), Devin and Gemini 3 Pro in a single command; ChatGPT Plus and OpenCode stage by stage; and GPT-6 Astra, ChatGPT Plus and OpenCode on a 14-variant verification task. They were compared with two conventional references, Galaxy and Google Colab.
- **Independence audit.** SHA-256 showed that 21 of 22 files in the first mission of OpenCode and ChatGPT Plus were byte-identical, so they were not independent runs and any agreement between them is not independent agreement.
- **Reading raw files reveals memory errors.** Gemini contradicted the raw GTEx file in 14 of 14 genes (16 of 16 contradictory claims; Wilson 80.6% to 100%) and attributed a non-existent classification to MFGA. OpenCode recomputed GTEx from the raw file and exposed the discrepancies. Devin detected an error in the NR5A1 representative, which was corrected and propagated.
- **Documenting tool limits.** GPT-6 Astra could not retrieve GTEx, OMIM and VarCards2 (0 of 14; Wilson 0 to 21.5%). This was documented as an execution-environment limitation, not a biological result, a distinction between “I did not find” and “it does not exist”.
- **Version instability.** Claude’s outputs changed between versions (14 vs 18 vs 15 in set M), so the version must be fixed with every result.
- **A proposed six-step protocol** for comparing AI tools in genomics: a unified candidate list, an explicit decision per tool, versions and fingerprints, independent environments, an independent reference standard, and pre-declared weights.
- The conclusion is that AI tools are useful for cross-checking when their outputs are verified against primary sources; they are not a source of truth.

13. Website and evidence database

The prototype. A bilingual (Arabic/English) prototype built with Streamlit, plus a static serverless version that runs offline without internet. It offers a chromosome map, gene cards and gene explorer, an evidence fingerprint, a structure gallery, tool comparison, a statistics workspace, database links, and a “selection lab” that re-applies the rule when the user changes thresholds. Its JavaScript engine recomputes Cochran’s Q, McNemar, Holm, Bonferroni, Wilson, Jaccard and bootstrap in the browser, with values matching the Python results. This turns the database from an archive into a selection lab whose claims the reader can test.

A multi-layer, multi-track evidence base. The database is a variant-level evidence matrix with independently sourced layers.

- **Genomic layer** (ClinVar at SCV level, CADD, gnomAD, 1000 Genomes): is the variant rare, how harmful is it, and for which disease was it registered?
- **Tissue-expression layer** (GTEx, HPA): is the gene expressed in the testis?
- **Functional layer** (SpliceAI, REVEL, AlphaMissense, LOFTEE, NMD): does the variant affect splicing or the protein?
- **Structural layer** (AlphaFold, pLDDT, ChimeraX, PyMOL, OMIM): where is the substitution, and does it fit the phenotype and mode of inheritance?

Why it is not an arbitrary database. A gene enters only with moderate or higher ClinGen evidence; a variant enters only if it is pathogenic or likely pathogenic with a submission linking it to the infertility phenotype; layers come from raw files, not summaries; and the database can be updated with each ClinVar release. It is described as evidence-based, reviewable and updatable, not definitively proven; no database in the world is 100% proven to cause infertility, and the project itself demonstrated this with its numbers.

Comparison with existing databases. Published male-infertility resources include the Male Fertility Gene Atlas, the Male Infertility Knowledgebase, IDDB, FertilityOnline and SpermatogenesisOnline, and some integrate tissue and testicular-cell expression and variant statistics. Adding expression layers is therefore not the new element. The unoccupied space, to our knowledge, is a variant-level interpretive database specialized in male infertility that matches disease at the SCV level, traces every value to its source and to more than one execution route, and lets the user re-apply the rule with custom thresholds. The current prototype is limited to this study’s results (16 genes) and is intended for later expansion.

14. Clinical path and practical output

- In the molecular-diagnosis gap, the project provides an interpretive layer that re-examines submitted variants and links them to the infertility phenotype; it complements clinical sequencing and does not replace it.
- If about three quarters of pathogenic variants in these genes are not registered for infertility, re-matching may prevent attributing a cancer variant or an androgen syndrome to infertility and redirect the case to the right specialty.
- The output is a disease-matched candidate list ready for functional validation. The distinction between “pathogenic in ClinVar” and “cause of infertility”, and between “candidate” and “diagnosis”, is preserved.

15. Strengths

- **Rigor:** thresholds fixed before seeing results; 15 representatives stable across nine scenarios.
- **Verification:** two independent routes matched exactly (469/469 CADD scores).
- **Traceability:** 13 evidence layers traced to raw files.
- **Self-correction:** raw-data recomputation fixed errors (for example MSH5 3.10 to 34.60), and the NR5A1 representative was corrected after Devin detected the error.
- **Honesty:** no AI difference survived Holm correction (smallest adjusted p = 0.1143).
- **Scale:** six chapters, 34 tables, 44 figures, more than 130 references, a 48-sheet Excel matrix and an audited log of 502 files.
- **Multiple skills in one coherent work:** bioinformatics, Python, inferential statistics, structural analysis, web development and AI-tool evaluation.
- **Scientific maturity:** the work criticizes itself, documents a substantive correction, discovers and corrects expression-value errors, and presents its limitations in a dedicated table.
- **Assessment:** defended with a grade of 97/100 and Honours.

16. Research in progress

Two papers derived from the project are in progress: (1) a reproducible submission-level disease-matching method for male-infertility genes; (2) a paired evaluation of AI tools against a verified reference in rare-disease variant filtering.

17. Limits

A candidate list, not a diagnosis.

18. Conclusion and strongest sentence

The project has a defensible novelty in a combination of elements: SCV-level disease matching, multiple evidence layers from raw files, a paired statistical comparison between bioinformatics and AI tracks with SHA-256 independence auditing, and an interactive website that recomputes the tests. These elements have not been combined in this way in male infertility, to our knowledge.

The strongest result in one sentence: about three quarters of variants classified as pathogenic in validated male-infertility genes are not registered for male infertility; “highest CADD” alone would select 6 of 15 representatives registered for other diseases; no AI difference survived Holm correction; and the tools that read raw files were the ones that caught the errors.

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